

AT32M412 BLDC Motor Control Library User Guide

Introduction

This application note describes how to drive a motor on the AT32M412_LV_MOTOR_DRV low-voltage motor evaluation board, using AT32 motor control library.

Applicable products

Product series	AT32M412 series
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1 Overview

Motor control library algorithms described hereinafter include:

Target motor type: Three-phase permanent magnet synchronous motor (BLDC)

Control methods

- 120° square wave commutation

Three-phase PWM method

- 120° conduction PWM control

Phase current sensing method

- 1-shunt current sensing and reconstruction method

Rotor position detection method

- Hall-effect position sensor

Initial rotor position estimation mode

- Three-phase voltage vector mode

Rotor position estimation in 120° square wave control mode

- BEMF zero-crossing point detection with ADC
- BEMF zero-crossing point detection using MCU's internal comparator

Sensored 120° square-wave commutation mode

- 120° square wave voltage control
- Torque control (120° square wave current control)
- Speed control
- Forward and reverse control

Sensorless 120° square-wave commutation mode

- 120° square wave voltage control
- Open loop startup
- Torque control (120° square-wave current control)
- Speed control

Auto tuning function

- Hall state sequence self-learning (for Hall sensor only)
- Motor winding parameters identification
- Current PI control parameters self-tuning

2 Environment requirements

2.1 Hardware environment

Hardware resources required include a PMSM (BLDC) motor, AT-Link or third-party programmer and a motor control evaluation board called AT32M412_LV_MOTOR_DRV. For more information on hardware configurations, please refer to the document titled “UM0016_AT32M412_LV_Motor_EVB_user_manual”.

- PMSM (BLDC) motor
- Debugger
- Motor control evaluation board: AT32M412_LV_MOTOR_DRV

Rated specifications for motor evaluation board

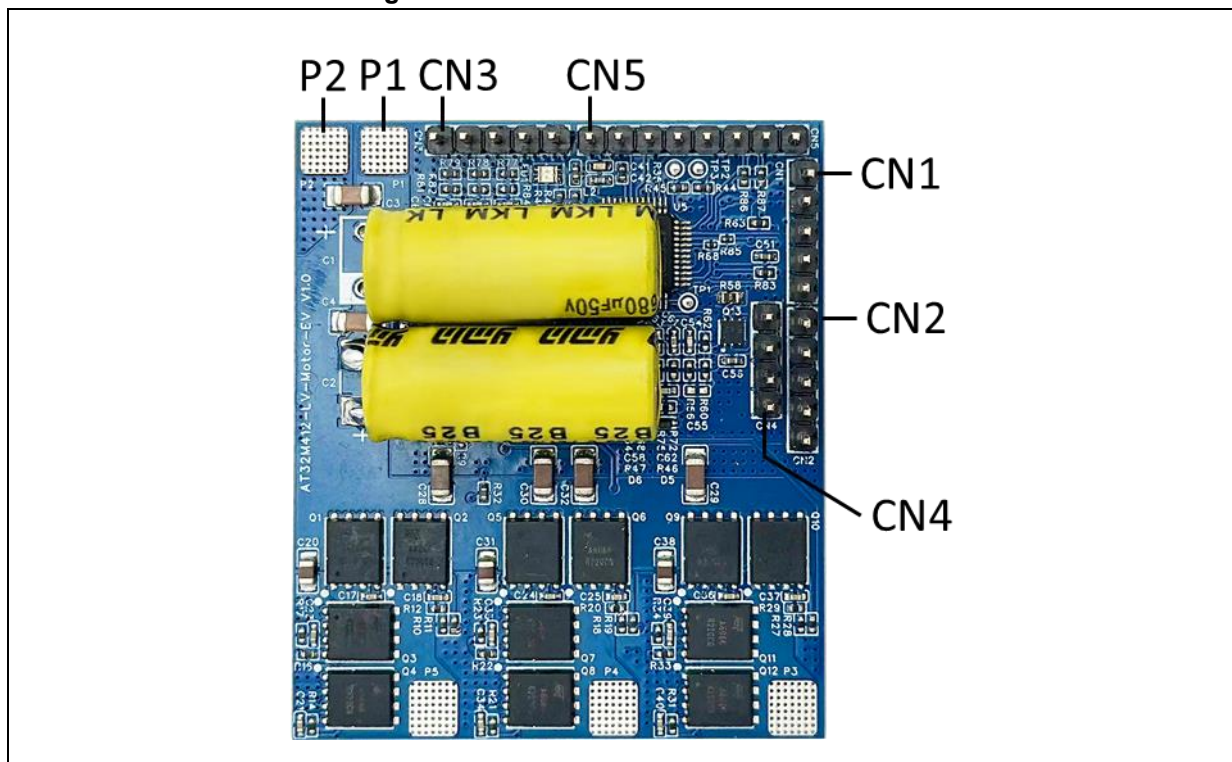
- Input voltage: 6V~45V
- Maximum output phase current: 45A_{PEAK}
- Overcurrent protection point: 55A_{PEAK}

Follow the procedures below to start with:

- 1) When using a position sensor, connect the Hall sensor (or incremental encoder) to CN3 on the board.
- 2) For external control input, connect a 0–3.3V analog signal to CN2 Pin 2 or a PWM input signal to CN2 Pin 3.
- 3) Connect AT-Link programmer (or third-party programmer) to CN1 on the board
- 4) Connect the TX/RX pins of the AT-Link programmer (or USB-to-TTL cable) to CN5 Pin 3 and Pin 4.
- 5) Download AT32M412 motor control demonstration code to the flash memory of MCU
- 6) Connect the motor's U, V, and W phases to P3, P4, and P5 on the board, respectively.
- 7) Adjust DC power supply voltage/current settings
- 8) Connect the positive terminal of the power supply (lithium polymer battery) to P1 and the negative terminal to P2 on the board, then power on.
- 9) Connect to the EVB board using ArteryMotorMonitor software. Clear the low-voltage error warning (caused by missing bus voltage after programming operation), set parameters, and control the motor operation.

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Figure 1. Connectors on evaluation board



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2.2 Software environment

- 1) Open “bldc_1shunt_sensorless” demo code
- 2) Run “ArteryMotorMonitor.exe” (no install is required)
- 3) If in Keil IDE environment, modify “Read/Only Memory Areas” according to actual AT32 MCU Flash memory size. See Table 1 for reference.

Example: for a 128-KB AT32M412CBT7, its IROM1 start address is set at 0x8000000, and length 0x1FC00. Its IROM2 start address is set at 0x801FC00, and the length 0x400, see Figure 2.

If in AT32 IDE environment, modify ID files according to actual AT32 MCU Flash memory size, see Figure 3.

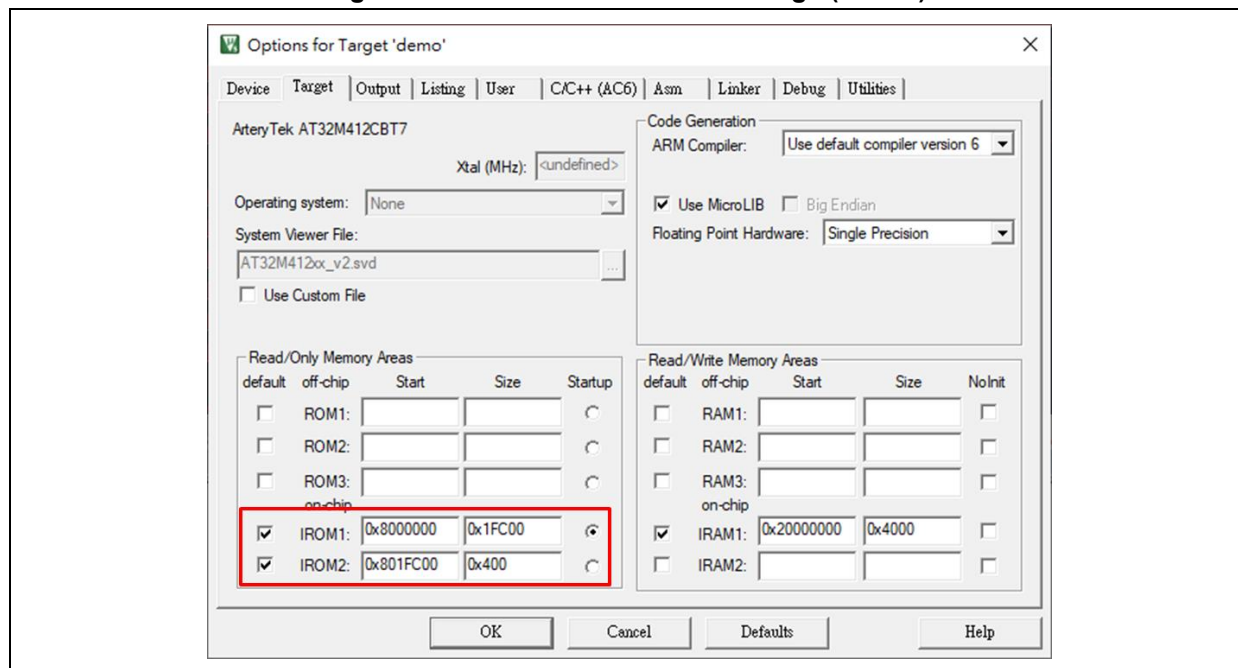
If in IAR environment, modify ICF files depending on actual AT32 MCU Flash memory size, see Figure 4 for reference.

Table 1. Correspondence between Flash size and ROM

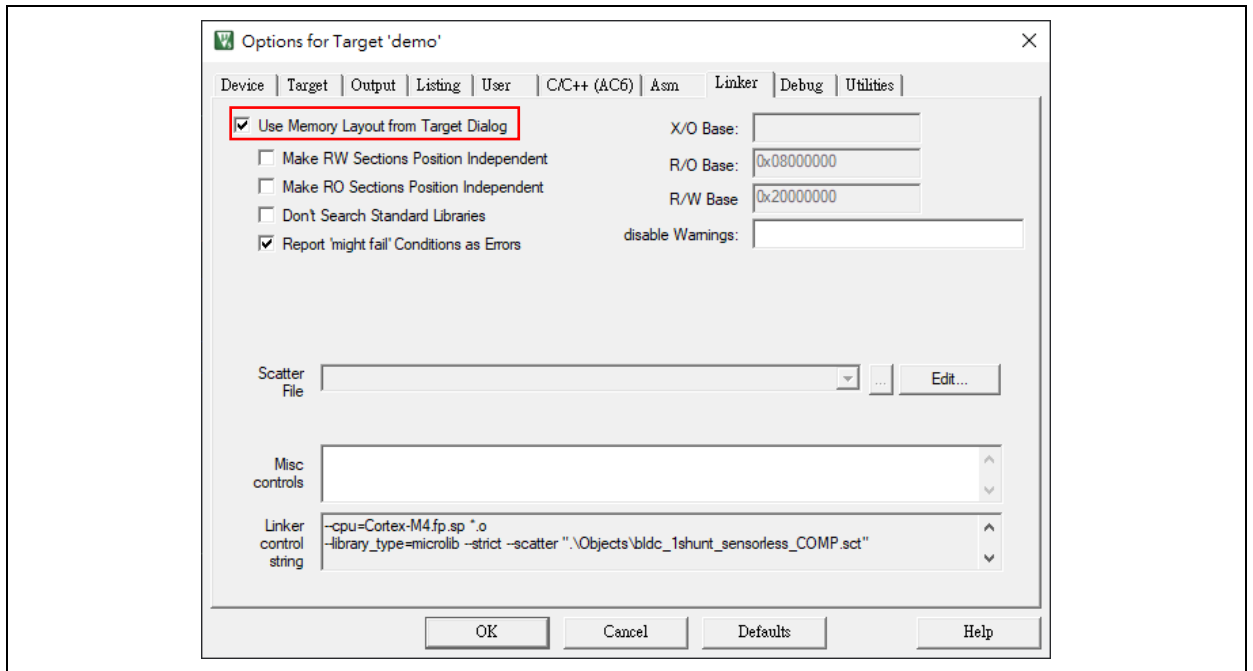
Flash size	128K	64K
IROM1(address)	0x8000000	0x8000000
IROM1(size)	0x1FC00	0x0FC00
IROM2(address)	0x801FC00	0x800FC00
IROM2(size)	0x400	0x400

Note [1]: Keil v5.33 (keil compiler version 5) is recommended as AT32 BSP source code does not support V6.15 compiler.

Figure 2. AT32M412CBT7 ROM settings (in Keil)



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Figure 3. AT32M412CBT7 ROM settings (in AT32IDE)

```
AT32M412x8_FLASH.ld
16  ** Environment : Arm gcc toolchain
17  **
18  ****
19  */
20
21  /* Entry Point */
22  ENTRY(Reset_Handler)
23
24  /* Highest address of the user mode stack */
25  _estack = 0x20004000; /* end of RAM */
26
27  /* Generate a link error if heap and stack don't fit into RAM */
28  _Min_Heap_Size = 0x200; /* required amount of heap */
29  _Min_Stack_Size = 0x400; /* required amount of stack */
30
31  /* Specify the memory areas */
32  MEMORY
33  {
34    FLASH (rx)      : ORIGIN = 0x08000000, LENGTH = 128K
35    MC_DATA (r)      : ORIGIN = 0x0801FC00, LENGTH = 0x400
36    RAM (xrw)       : ORIGIN = 0x20000000, LENGTH = 16K
37  }
38
39  /* Define output sections */
40  SECTIONS
41  {
42    /* The startup code goes first into FLASH */
43    .isr_vector :
44    {
45      . = ALIGN(4);
46      KEEP(*(.isr_vector)) /* Startup code */
47      . = ALIGN(4);
48    } >FLASH
49
50    .mc_data :
51    {
52      . = ALIGN(4);
53      . = ORIGIN(MC_DATA);
54      _MC_VectStoreAddr1 = .;
55      *(.MC_VectStoreAddr1);
56      . = ALIGN(4);
57      . = ORIGIN(MC_DATA) + 800;
58      _MC_VectStoreAddr2 = .;
59      *(.MC_VectStoreAddr2);
60      . = ALIGN(4);
61    } >MC_DATA
```

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Figure 4. AT32M412CBT7 ROM settings (in IAR)

```
AT32M412x8_mcddata.icf
1  /****ICF**** Section handled by ICF editor, don't touch! ****/
2  /*-Editor annotation file-*/
3  /* IcfEditorFile="$TOOLKIT_DIR$\config\ide\IcfEditor\cortex_v1_0.xml" */
4  /*-Specials-*/
5  define symbol __ICFEDIT_intvec_start__ = 0x08000000;
6  /*-Memory Regions-*/
7  define symbol __ICFEDIT_region_ROM_start__ = 0x08000000;
8  define symbol __ICFEDIT_region_ROM_end__ = 0x0801FC00 - 1;
9  define symbol __ICFEDIT_region_ROM1_start__ = 0x0801FC00;
10 define symbol __ICFEDIT_region_ROM1_end__ = (__ICFEDIT_region_ROM1_start__ + 800 - 1);
11 define symbol __ICFEDIT_region_ROM2_start__ = (__ICFEDIT_region_ROM1_end__ + 1);
12 define symbol __ICFEDIT_region_ROM2_end__ = 0x0801FC00 + 0x400 - 1;
13 define symbol __ICFEDIT_region_RAM_start__ = 0x20000000;
14 define symbol __ICFEDIT_region_RAM_end__ = 0x20003FFF;
15 /*-Sizes-*/
16 define symbol __ICFEDIT_size_cstack__ = 0x400;
17 define symbol __ICFEDIT_size_heap__ = 0x200;
18 /**** End of ICF editor section. ****ICF****/
19
20 define memory mem with size = 4G;
21 define region ROM_region = mem:[from __ICFEDIT_region_ROM_start__ to __ICFEDIT_region_ROM_end__];
22 define region ROM1_region = mem:[from __ICFEDIT_region_ROM1_start__ to __ICFEDIT_region_ROM1_end__];
23 define region ROM2_region = mem:[from __ICFEDIT_region_ROM2_start__ to __ICFEDIT_region_ROM2_end__];
24 define region RAM_region = mem:[from __ICFEDIT_region_RAM_start__ to __ICFEDIT_region_RAM_end__];
25
26 define block CSTACK with alignment = 8, size = __ICFEDIT_size_cstack__ { };
27 define block HEAP with alignment = 8, size = __ICFEDIT_size_heap__ { };
28
29 initialize by copy { readwrite };
30 do not initialize { section .noinit };
31
32 place at address mem: __ICFEDIT_intvec_start__ { readonly section .intvec };
33
34
35 place in ROM_region { readonly };
36 place in ROM1_region { readonly section .MC_VectStoreAddr1 };
37 place in ROM2_region { readonly section .MC_VectStoreAddr2 };
38 place in RAM_region { readwrite,
39 block CSTACK, block HEAP };
```

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3 BLDC motor control library files

References are made to the following documents:

Hardware circuit of the AT32M412 low-voltage motor control evaluation board:

UM0016_AT32M412_LV_Moto_EVB_V1_User_Manual

Program file structure, macro definition and state machine flow chart:

AN0169_AT32_BLDC_Hall_1-shunt_Demo

AN0170_AT32_BLDC_Sensorless_1-shunt_Demo

Motor control library functions:

AN0064_AT32_Motor_Control_Library_User_Guide

Motor control fine-tuning procedures:

AN0213_AT32_BLDC_Hall_Quick_Start_Guide

AN0214_AT32_BLDC_Sensorless_Quick_Start_Guide

Host computer control program: AN0063_AT32_Motor Monitor_Application_Note

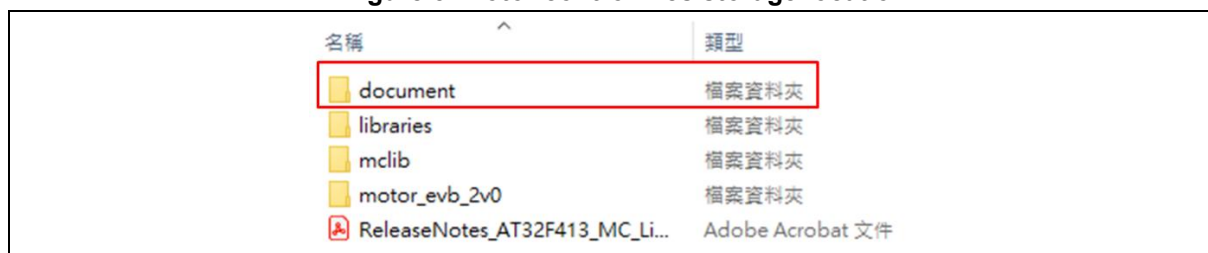
Host computer software transmission protocol: AN0199_AT32_Motor Monitor_Protocol

Table 2. List of reference documents

Description	Reference document
1. Hardware circuit	UM0016_AT32M412_LV_Moto_EVB_V1_User_Manual
2. Program files structure	AN0169_AT32_BLDC_Hall_1-shunt_Demo
3. Macro definition	
4. State machine flowchart	AN0170_AT32_BLDC_Sensorless_1-shunt_Demo
5. Motor control library functions	AN0064_AT32_Motor_Control_Library_User_Guide
6. Motor control fine-tuning procedure	AN0213_AT32_BLDC_Hall_Quick_Start_Guide
	AN0214_AT32_BLDC_Sensorless_Quick_Start_Guide
7. Host computer control program	AN0063_AT32_Motor Monitor_Application_Note
8. Host computer transmission protocol	AN0199_AT32_Motor Monitor_Protocol

All these documents are available at ARTERY official website. AN0064, AN0169, AN0170, AN0171, AN0213 and AN0214 are packed into a motor control library folder in .zip format. AN0063 and AN0199 are packed into a host computer folder in .zip format

Figure 5. Motor control files storage location



名稱	類型
document	檔案資料夾
libraries	檔案資料夾
mclib	檔案資料夾
motor_evb_2v0	檔案資料夾
ReleaseNotes_AT32F413_MC_Li...	Adobe Acrobat 文件

4 Quick start guide

4.1 Six-step square wave sensed control

For details on how to fine-tune BLDC motor using six-step square wave, refer to the document “AN0213_AT32_BLDC_Hall_Quick_Start_Guide”.

For AT32M412 low-voltage motor control evaluation board, one of its distinctive features lies in the use of internal comparator and DAC for protection against overcurrent. See Section 4.4 for details.

4.2 Six-step square wave sensorless control

For details on how to fine-tune BLDC motor using six-step square wave sensorless method, refer to the document “AN0214_AT32_BLDC_Sensorless_Quick_Start_Guide”.

In sensorless control mode, the commutation timing point is estimated via BEMF. It is therefore possible to use either ADC or comparator to detect a zero-crossing point. On the comparator part, it can be an external comparator or an internal one. In the case of the AT32M412 and AT32M416, they both come with internal comparators, so our example use an internal comparator to demonstrate how to detect a zero-crossing point.

In addition, when using an internal comparator for zero-crossing point detection, it is required to adjust sensorless parameters before fine-tuning blanking window. See Section 4.5 for details.

For AT32M412 low-voltage motor control evaluation board, one of its distinctive features lies in the use of internal comparator and DAC for protection against overcurrent. See Section 4.4 for details.

4.3 Current sensing

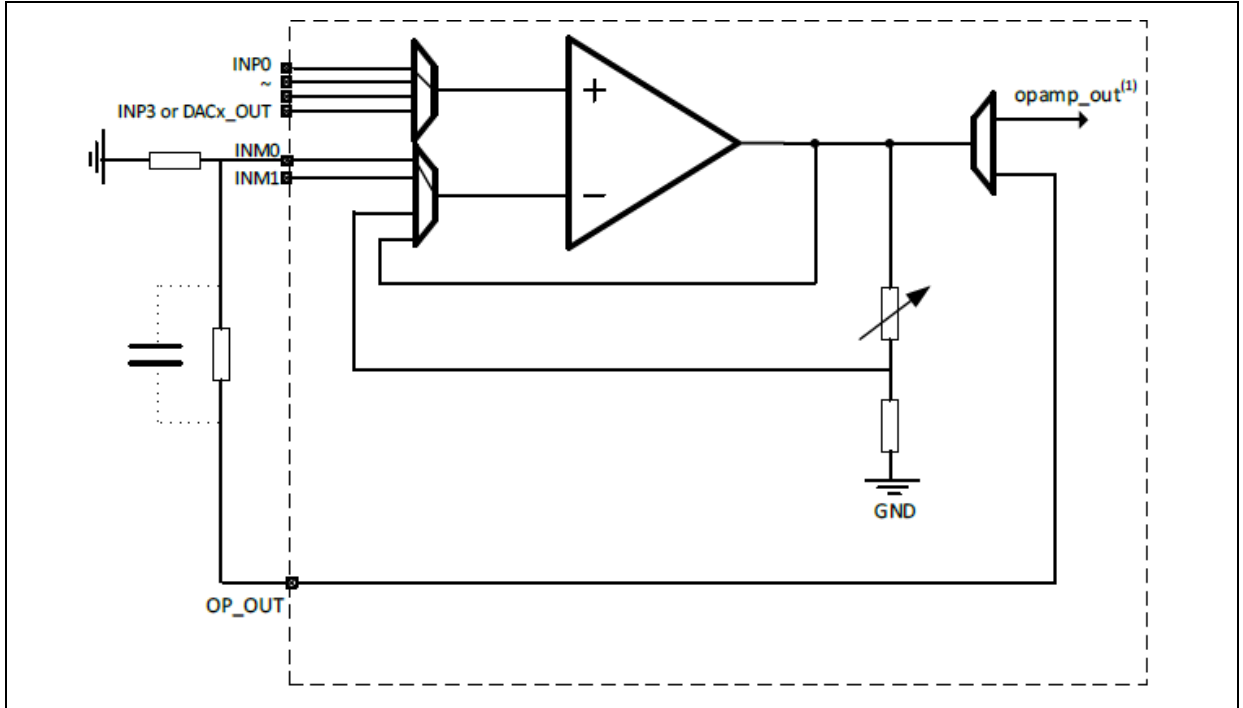
The AT32M412 series and the AT32M416 series are equipped with four operational amplifiers (OP), which can be used for sensing BLDC current. Our example case uses an OP in Standalone mode to detect bus current signals.

Follow the procedure below to enter standalone mode:

- Set OPx_CTRL.INMSEL[1:0] = 0x
- Configure corresponding INP/INM channels as analog mode
- If OPx_CTRL.OPFSTMD=1 is asserted, the OP enters fast mode
- Set OPx_CTRL.OPEN to 1 to enable Standalone mode.

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Figure 6. Circuit diagram in Standalone mode

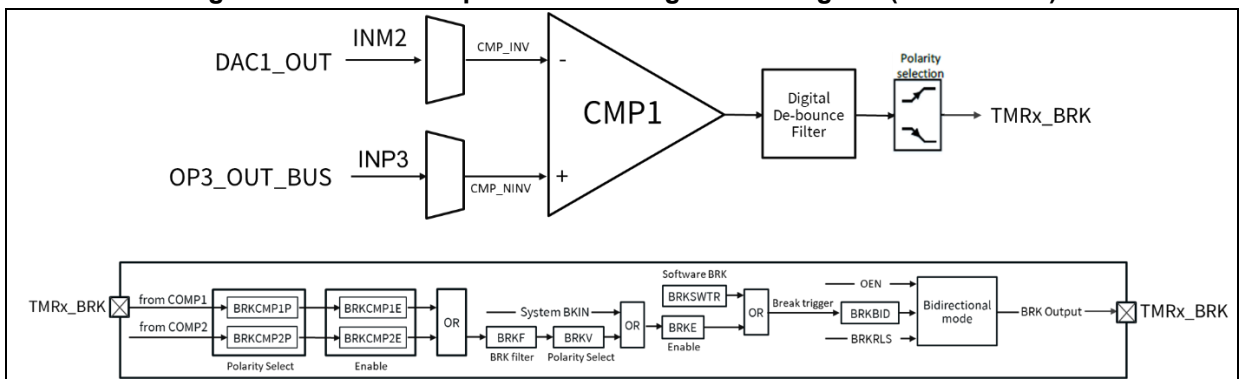


4.4 Current protection

The AT32M412 series and AT32M416 series are equipped with DAC, comparators and brake input feature, which enable them to monitor overcurrent phenomenon.

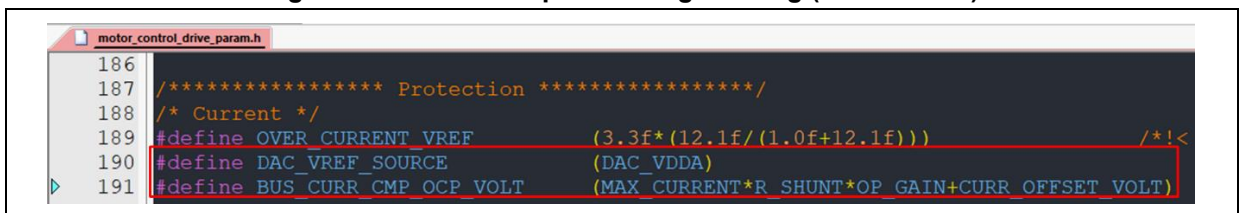
The DAC output voltage acts as an overcurrent protection point that is compared with current input signals, with the comparison result as the input signal of the TMRx_BRK, as shown in Figure 7.

Figure 7. Overcurrent protection configuration diagram (M412 series)



The voltage of the overcurrent protection point is configurable by modifying macro definition, and the reference voltage source for DAC is also selectable, as shown in Figure 8.

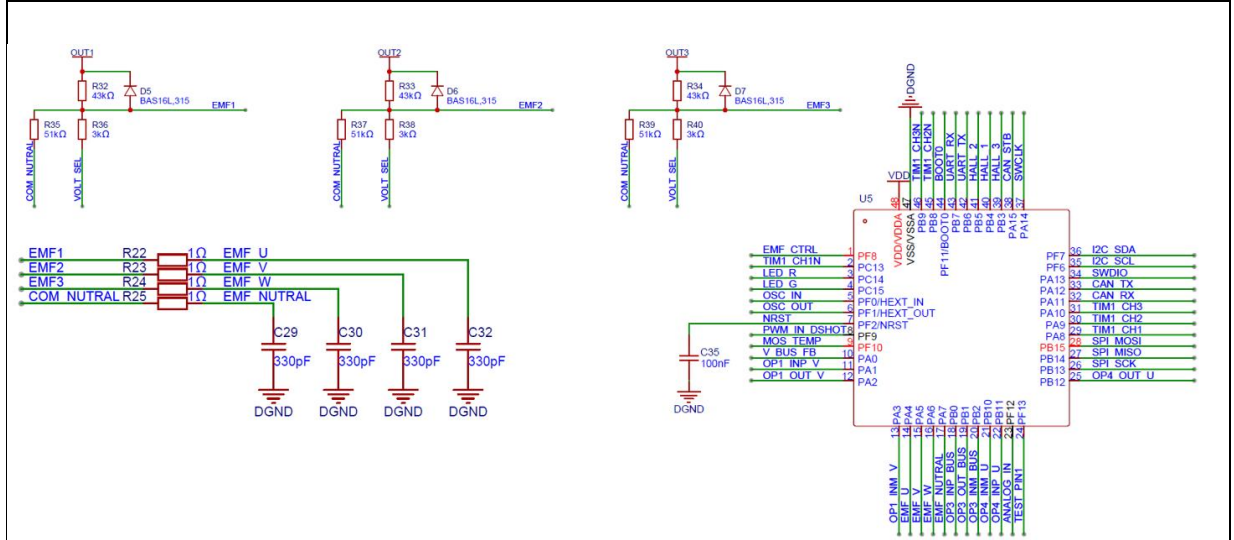
Figure 8. Overcurrent point voltage setting (M412 series)



4.5 BEMF zero-crossing detection with comparator

As mentioned before, the BEMF zero-crossing point can also be monitored using a comparator. As shown in Figure 9, the zero-crossing point is obtained by comparing a three-phase BEMF divided voltage signal with a synthesized neutral point.

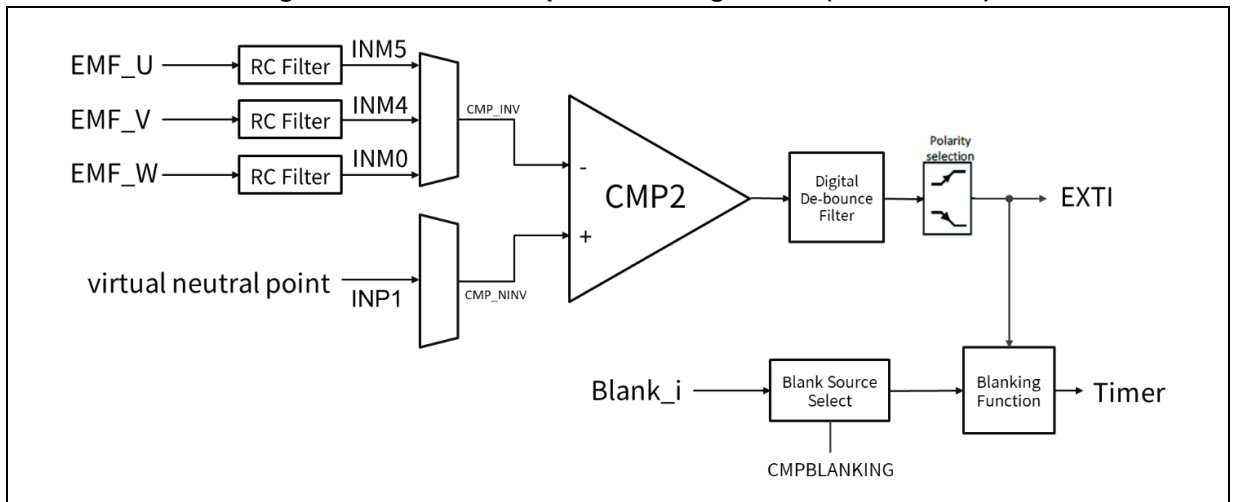
Figure 9. BEMF voltage divider circuit with comparator (M412 series)



When it comes to BEMF zero-crossing point detection, it can be done with an on-chip or external comparator.

In order to save costs and reduce PCB footprint, ARTERY offers a rich selection of MCUs featuring internal comparators, such as AT32F421, AT32F415, AT32M412 and AT32M416. See Figure 10 for configuration details.

Figure 10. Internal comparator configuration (M412 series)



When motor is in running mode, there is some potential noise as a result of switching on/off MOSFET, which can be filtered out by enabling the blanking window of the comparator for better performance. In addition, if the blanking window is properly configured, the compared signal can be directly connected to a timer capture channel, triggering a 30-degree phase delay for commutation. This will help save the need of filtering noise with software and thus achieve a quicker commutation timing.

This method is already included in the example case of sensorless BLDC motor control. Thus it is possible to adjust blanking window size by modifying the EMF_RISE_BLANK_TIME and EMF_FALL_BLANK_TIME.

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Figure 11 shows how to configure blanking window size.

Figure 11. Blanking window configuration

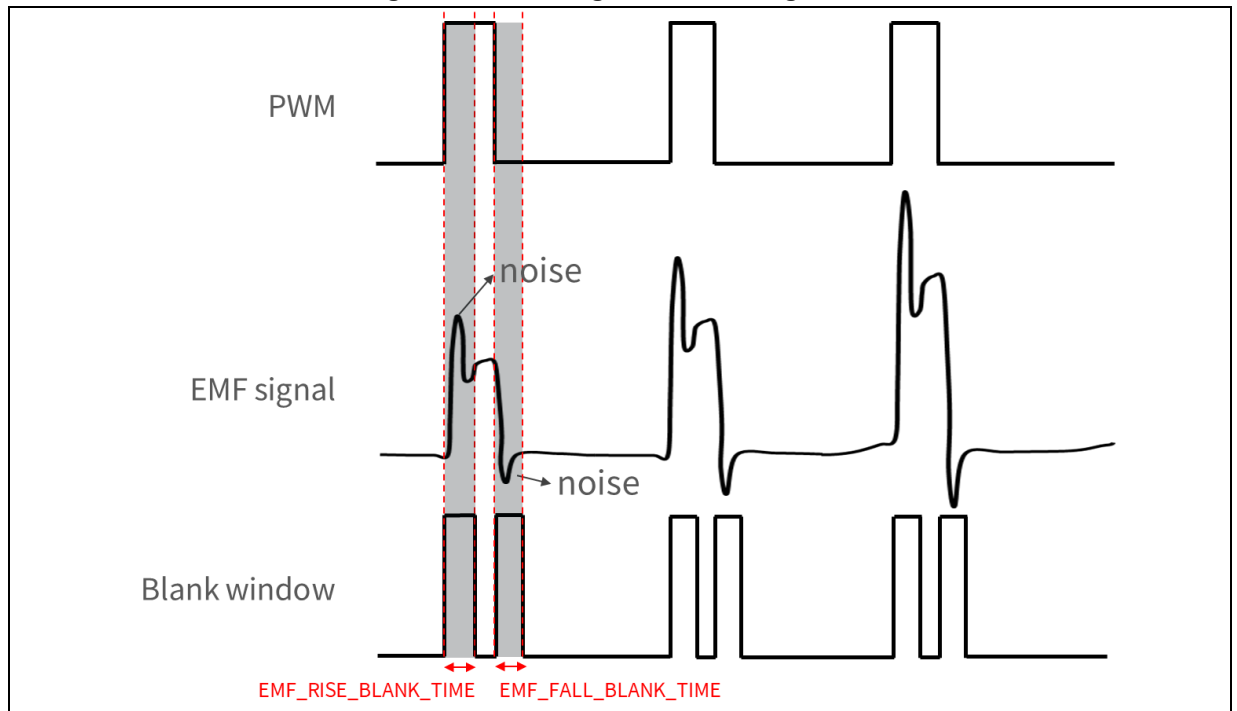


Figure 12. Blanking window size definition

```
motor_control_drive_param.h
164
165 /* EMF */
166 #define EMF_SENSE_GAIN          (3.0f/43.0f+3.0f)    /*!< 0.0652 */
167 #define GATE_DELAY_TIME        (0.8)                /*!< usec */
168 #define EMF_SIG_RISING_TIME    (9.8)                /*!< usec */
169 #define EMF_SIG_FALLING_TIME   (21.5)               /*!< usec */
170
171 #define EMF_RISE_BLANK_TIME     (30.0)               /*!< usec */
172 #define EMF_FALL_BLANK_TIME    (20.0)               /*!< usec */
173 #define BLANK_WINDOW_INIT      (240)
174 #define EMF_MIN_VALUE          (90)
175
```


5 Revision history

Table 3. Document revision history

Date	Revision	Changes
2025.08.28	1.0.0	Initial release

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